# THE ROTATED ADMISSIBLE TRIANGLE IS ISOMORPHIC TO THE CONE TRIANGLE OF THE ROTATED DIFFERENTIAL

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the splitting data and the two comparison maps, the differential closedness computations with the d(tilde-alpha) and d(tilde-p) identities, the diagram fit, and both composition identities collapsing to the respective identity morphisms are all retained verbatim (the author marks the diagram fit as immediate; the closedness and composition computations that carry the proof are displayed in full).

*Editorial note.* The label lemma-cone-rotate-isom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Identities for the splitting data of an admissible sequence. Same chapter, Lemma lemma-get-triangle. Supplies the relations delta[-1] = tilde-alpha d(tilde-beta) = -d(tilde-alpha) tilde-beta and delta[-1] = tilde-i d(tilde-p) together with equation (6), which drive every differential check and the composition computation.
- Cones of morphisms in the differential graded category. Same chapter, Definition definition-cone and Situation situation-ABC. Fixes the cone c(delta[-1]) with its structure maps i and p whose triangle is compared with the rotated admissible triangle.

*Source reference.* Original reference 09P6: sources/stacks-dga/pinned-original-source.tex, source lines 3708–3708. Prerequisite pointer invoked at proof line(s) 4207; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09QJ: sources/stacks-dga/pinned-original-source.tex, source lines 3762–3762. Prerequisite pointer invoked at proof line(s) 4168; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 09QJ let $\alpha : x \to y$ and $\beta : y \to z$ define an admissible short exact sequence*

$$x \longrightarrow y \longrightarrow z$$

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*in Comp($\mathcal{A}$). Let $(x, y, z, \alpha, \beta, \delta)$ be the associated triangle in $K(\mathcal{A})$. Then, the triangles*

$$(z[-1], x, y, \delta[-1], \alpha, \beta) \quad \textit{and} \quad (z[-1], x, c(\delta[-1]), \delta[-1], i, p)$$

*are isomorphic.*

*Proof.* We have a diagram of the form

$$\begin{array}{ccccccc}
z[-1] & \xrightarrow{\delta[-1]} & x & \underset{\tilde{\alpha}}{\overset{\alpha}{\rightleftarrows}} & y & \underset{\tilde{\beta}}{\overset{\beta}{\rightleftarrows}} & z \\
\downarrow 1 & & \downarrow 1 & & \downarrow & & \downarrow 1 \\
z[-1] & \xrightarrow{\delta[-1]} & x & \underset{\tilde{i}}{\overset{i}{\rightleftarrows}} & c(\delta[-1]) & \underset{\tilde{p}}{\overset{p}{\rightleftarrows}} & z
\end{array}$$

with splittings to $\alpha, \beta, i$, and $p$ given by $\tilde{\alpha}, \tilde{\beta}, \tilde{i}$, and $\tilde{p}$ respectively. Define a morphism $y \to c(\delta[-1])$ by $i\tilde{\alpha} + \tilde{p}\beta$ and a morphism $c(\delta[-1]) \to y$ by $\alpha\tilde{i} + \tilde{\beta}p$. Let us first check that these define morphisms in Comp($\mathcal{A}$). We remark that by identities from Lemma 09P6, we have the relation $\delta[-1] = \tilde{\alpha}d(\tilde{\beta}) = -d(\tilde{\alpha})\tilde{\beta}$ and the relation $\delta[-1] = \tilde{i}d(\tilde{p})$. Then

$$\begin{aligned}
d(\tilde{\alpha}) &= d(\tilde{\alpha})\tilde{\beta}\beta \\
&= -\delta[-1]\beta
\end{aligned}$$

where we have used equation (6) of Lemma 09P6 for the first equality and the preceding remark for the second. Similarly, we obtain $d(\tilde{p}) = i\delta[-1]$. Hence

$$\begin{aligned}
d(i\tilde{\alpha} + \tilde{p}\beta) &= d(i)\tilde{\alpha} + id(\tilde{\alpha}) + d(\tilde{p})\beta + \tilde{p}d(\beta) \\
&= id(\tilde{\alpha}) + d(\tilde{p})\beta \\
&= -i\delta[-1]\beta + i\delta[-1]\beta \\
&= 0
\end{aligned}$$

so $i\tilde{\alpha} + \tilde{p}\beta$ is indeed a morphism of Comp($\mathcal{A}$). By a similar calculation, $\alpha\tilde{i} + \tilde{\beta}p$ is also a morphism of Comp($\mathcal{A}$). It is immediate that these morphisms fit in the commutative diagram. We compute:

$$\begin{aligned}
(i\tilde{\alpha} + \tilde{p}\beta)(\alpha\tilde{i} + \tilde{\beta}p) &= i\tilde{\alpha}\alpha\tilde{i} + i\tilde{\alpha}\tilde{\beta}p + \tilde{p}\beta\alpha\tilde{i} + \tilde{p}\beta\tilde{\beta}p \\
&= i\tilde{i} + \tilde{p}p \\
&= 1_{c(\delta[-1])}
\end{aligned}$$

where we have freely used the identities of Lemma 09P6. Similarly, we compute $(\alpha\tilde{i} + \tilde{\beta}p)(i\tilde{\alpha} + \tilde{p}\beta) = 1_y$, so we conclude $y \cong c(\delta[-1])$. Hence, the two triangles in question are isomorphic. □